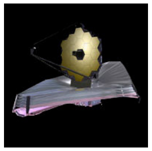

3273 - Eureka!: An Open-Source Pipeline for JWST Time-Series Observations

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The reduction of JWST time-series observations (TSOs) of transiting exoplanets requires careful analysis using customized software that is not provided as part of STScI's official "jwst"' software package. These custom data reduction pipelines are seldom open-source and oftentimes lack documentation, thorough testing, and general user support. This creates a members-only environment for those that have access to one of the few validated pipelines. Without access to or training for such software, new researchers trying to enter the field face a daunting barrier.

The philosophy behind the Eureka! project is to facilitate a community-supported, open-source pipeline that is modular in design and easy to use. The Eureka! pipeline has been an unmitigated success, with a handful of publications in the first six months of JWST science operations. We propose to build on this momentum by (1) adding support for additional TSO instrument modes, (2) enhancing the light curve fitting stage, (3) incorporating new and improved algorithms, and (4) addressing a growing list of unresolved issues and suggested enhancements brought on by people's experiences with Cycle 1 data. The ultimate goal is to build a robust and stable end-to-end pipeline that will service the transiting exoplanet and brown dwarf communities over the next decade of JWST science operations.

OBSERVING DESCRIPTION

This is an archival Community Data Science Software proposal.

To date, funding for Eureka! has been cobbled together through a haphazard collection of sources (e.g., ERS, GTO, and student/postdoc funding). These funding sources have mostly run out, leaving Eureka! in a state of uncertainty. JWST GO programs do not provide sufficient financial support to actively maintain software packages, let alone realize our four primary objectives. The Community Data Science Software initiative is the best opportunity to make sure Eureka! is maintained in the long term, and the JWST} community has access to effective, open-source, and user-friendly data reduction and analysis code.

We propose to build upon Eureka!'s successful origins by accomplishing four primary objectives:
(1) Adding support for additional TSO instrument modes,
(2) Enhancing the light curve fitting stage,
(3) Incorporating new and improved algorithms, and
(4) Addressing a growing list of unresolved issues and suggested enhancements brought on by people's experiences with Cycle 1 data.